

1. (6 pts) The matrix $A = \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 2 & 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & -15 & -1 \\ 1 & 0 & 2 & 4 & 2 \end{bmatrix}$ row reduces to $\begin{bmatrix} 1 & 0 & 0 & -34 & -4 \\ 0 & 1 & 0 & 24 & 3 \\ 0 & 0 & 1 & 19 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) Find a basis for the column space of A .

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

$$x_1 = 34x_4 + 4x_5$$

$$x_2 = -24x_4 - 3x_5$$

$$x_3 = -19x_4 - 3x_5$$

$$x_4 = 1x_4 + 0x_5$$

$$x_5 = 0x_4 + 1x_5$$

b) Find a basis for the null space of A .

$$\left\{ \begin{bmatrix} 34 \\ -24 \\ -19 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ -3 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

2. (6 pts) Let A be a 6×5 matrix. Determine if each statement is True or False and give a brief explanation of your answer.

a) If the rank of A is 5 then $\text{Col } A = \mathbb{R}^5$.

False $\text{Col } A$ is a 5 dim'l subspace of $\mathbb{R}^6$

b) If the rank of A is 5 then $\text{Nul } A = \{\mathbf{0}\}$.

True Rank = 5 means a pivot in every column so there are no free variables.
This means the homogeneous system $A\vec{x} = \vec{0}$ has the unique solution $\vec{0}$.

3. (3 pts) List three things that are true about a matrix A if the columns of A form a basis for $\mathbb{R}^3$.

Many choices are possible

$\text{Col } A = \mathbb{R}^3$, $\text{Nul } A = \{\mathbf{0}\}$, columns are linearly independent

A has 3 rows and 3 columns, A is invertible

$\det A \neq 0$, A has pivots in every row, pivots in every column